What data the Pulse SDK collects

- The SDK records **what visitors do** (pages, clicks, which fields they enter) and **what breaks** (errors, failed requests, buttons that do nothing).
- It sends field **labels, never typed values**, and **removes `?query` and `#hash`** from URLs.
- Visitors' own words are sent in only 3 places: **survey answers**, **search-box text** and the **"I'm lost" guide**.
- No cookies, no IP address, no location permission, no screen recording.

1 What is sent, and where it goes

Request	What it carries	Stored in
`POST /v1/events`	Browser info + batches of `interaction` and `friction` events. Sent every 5 s, at 20 events, or when the tab closes.	`sessions`, `interaction_events`, `friction_events`
`POST /v1/responses`	A survey answer, or that it was closed	`survey_responses`
`POST /v1/surveys/generate`	One friction event + browser summary, so AI can write a survey (only if the site turns AI on)	`generated_surveys`, and sent to **OpenAI** (minus `stack`)
`POST /v1/guide`	What the visitor typed into "I'm lost"	Not stored. Sent to **OpenAI** only when the server has an OpenAI key (to find pages by meaning)
`GET /v1/surveys`	Only the site key	–

Every POST also has `siteKey` (public site id) and `sessionId` (random id for the tab, erased when it closes).

2 Every field

How to read it: `red` = contains personal info (**YES**). `orange` = usually safe, but can carry personal info on some sites (**MAYBE**). Plain = cannot (numbers, fixed words, ids we make up).

Group	Fields
Browser info (sent with each event batch)	
`meta`	`userAgent`, `language`, `viewport` (window size), `referrer`
`meta.client`	`timezone`, `region` (city name from the timezone, no GPS), `localTime`, `partOfDay`, `weekday`, `language`, `browser`, `os`, `device`, `viewport`, `touch`, `connection`, `referrer`
Interaction events: one per step the visitor takes	
every event	`kind`, `type`, `at` (time), `url`
`page_view`	`title` (page title)
`click`	`selector`, `tag`, `label` (button text), `area` (heading of its section), `field` (label of the box next to it), `x`, `y`, `disabled`
`focus`	`field` (the field's label, not its value), `selector`
`submit`	`selector`
`error`	`kind`, `message`, `status`
Friction events: one per moment the visitor got stuck	
every event	`kind`, `id`, `type`, `at`, `url`, `title`, `eligible`, `recordOnly`, `element` (`selector`, `tag`, `label`, `area`, `field`), `trail` (last 50 interaction events)
`rage_click`	`clickCount`, `durationMs`, `disabled`, `score`, `signals`
`dead_click`	`reason`, `waitedMs`, `score`, `signals`
`form_abandon`	`reason`, `idleMs`, `fieldsTouched`, `fieldsTotal`, `requiredRemaining`, `lastField` (label), `lastFieldSelector`

Group	Fields
`lost`	`tier`, `score`, `strong`, `patternIds`, `query` (last text typed in a search box, max 80 chars)
`manual`	`source`, `intent` (`lost`, `issue` or `question`)
Friction `error`: every kind also has `causedByClick`, `score`, `signals`	
`page` (404/500)	`status`, `message`, `cameFrom`, `cameFromTitle`
`js` (crash)	`message`, `source`, `line`, `column`, `stack`
`promise`	`message`, `stack`
`console`	`message` (whatever the site logged)
`resource`	`message`, `resource` (file URL)
`network` (5xx)	`method`, `url`, `status`, `message`
Other requests	
survey answer	`triggerId`, `templateId` or `generatedSurveyId`, `triggerType`, `status`, `autoOpened`, `url`, `answers` (free text is exactly what was typed, max 2000 chars)
guide	`query` (max 200 chars), `url`

Example (real click from Skyline Air):

```
{kind:"interaction", type:"click", url:"http://localhost:5173/checkout.html",
 data:{selector:"#apply-promo", tag:"button", label:"Apply", area:"Passenger details",
       field:"Promo code", x:618, y:444, disabled:false}}
```

3 What is NOT collected

- **Typed form values.** Only the field's label. *Tested:* typed `SECRETVALUE leak@example.com` into every field on all 5 sites. It was stored only where expected (search-box `query`, survey answers).
- **Password and email boxes**: never read, even if they look like a search box.
- **Query strings and #hashes** in page, referrer, file and request URLs. *Tested:* loaded every page with `?q=leak@example.com` and `#token=abc`. No stored URL kept them. (Exception: `stack`, see below.)
- **Request and response bodies** of the site's API calls (only method, URL and status).
- **Page content, screenshots, mouse movement, keystrokes, cookies, IP address, logins or user ids.**

4 Where personal info can slip through

	Field	Risk
1	Survey `answers` (free text)	Visitors type anything, e.g. "my email is leak@example.com" (stored as-is on all 5 sites).
2	Search `query` (`lost`), guide `query`	Search text often has names or emails. Seen on Harbor: `SECRETVALUE leak@example.com` stored, and sent to OpenAI when AI is on. The guide `query` is not stored, but it is sent to OpenAI when the server has a key.
3	`stack` (`js`, `promise`)	**Bug:** not cleaned, so it keeps the query string. Stored: `.../checkout.html?q=leak@example.com:69:...`
4	Error `message`	Sites sometimes log user data in errors, e.g. `console.error("failed for", email)`.
5	`label`, `area`, `title`, URL paths	Visible text or paths can include a name ("Hi, Jane", `/users/jane-doe`).
6	`userAgent` + timezone + window size	Together they can help fingerprint a browser.

5 How this was tested

1. Read every place the SDK sends data, and what the server stores and sends to OpenAI.

2. Drove all 5 demo sites (Skyline Air, Northline, Harbor, Westbridge, Linden) in a browser on a Docker image rebuilt from the latest push. On each: visited every page, typed into every field, clicked and rage-clicked buttons, scrolled, forced a JS crash, a failed promise, a `console.error`, a missing image and a failed request, opened a 404, answered a survey and used the guide.
3. Checked what the database stored. All 5 interaction types, all 6 friction types and all 6 error kinds appeared. Every stored field is in the table above.

Note: only Skyline Air's 404 page loads the SDK, so `page` errors were only captured there.